

RONALD GARCIA

rgarci47@jh.edu — (786) 298-4231 — github.com/Ronald-Garcia — United States

EDUCATION

Johns Hopkins University, Baltimore, MD
BS Mechanical, Computer, & Electrical Engineering

August 2022 — Anticipated: May 2026
Overall GPA: 3.93

EXPERIENCE

WSE Manufacturing
Course TA

Baltimore, MD
August 2024 – Now

- Collaborated with machinists and lecturers to facilitate the Manufacturing course at Johns Hopkins University.
- Prepared redesigned Solidworks assembly drawings, exploded views
- Machined Stirling engine and demonstrated tips and tricks to students

Manufacturing Student Tech
TA

Baltimore, MD
October 2023 - Today

- Work with Manufacturing engineers/machinists to design solutions for clients, ranging from Johns Hopkins Medical Campus to local startups.
- Designed parts in SolidWorks for clients, reworked identification system in C++, trained in the Manufacturing Student shop, and worked in a team of engineers to ideate and produce prototypes.

PROJECTS

The Mechanically-Chargeable Car Race
August 2022 - December 2022

Baltimore, MD

- The Mechanical Engineering Department at Johns Hopkins University hosted a race where teams design a small wheeled vehicle that is able to:
 - Receive the energy from a 5-pound weight falling 5 feet.
 - Store the energy and
 - Release and use the energy to go forward 25 feet.
- Worked in a team, aided with laser cutting and designing the hull, free-wheel mechanism, and CAD models of the vehicle.

The Piano Game
January 2023 - May 2023

Baltimore, MD

- The Mechanical Engineering Department at Johns Hopkins University hosted an exhibition where teams design a carnival game that teaches the user about a certain country.
- Worked in a team, programmed the game in Arduino.
- Aided in the CAD models and the manufacturing of the hull using jigsaw and laser cutter.

Mechanically Powered Crane
January 2024 - May 2024

Baltimore, MD

- CADed entire assembly for machined crane.
- Calculated mechanical requirements for gearbox in a team.
- Designed gearbox and aided in assembling crane.

SKILLS

- **Relevant Coursework:** Mastering Electronics I & the Lab, Intermediate Programming, Statics & Mechanics of Materials, Manufacturing Engineering, Mechanics Based Design, MechE Dynamics, MechE Thermodynamics, Materials Selection, Fluid Dynamics.
- **Programming:** C++, Java, C, Python, MATLAB, C#, Javascript, HTML, CSS, Typescript, REACT.js.
- **Software:** JupyterLab, Solidworks, LTSpice, MATLAB Simulink, Visual Studio Code, Stratasys, Overleaf.
- **Communication:** Leadership, over 8 years of tutoring/teaching experience.